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3. Hydrogen peroxide is a product of respiration and is made in all cells. It is harmful and therefore removed as soon as it is produced. Catalase is an intracellular enzyme that converts hydrogen peroxide into water and oxygen.

(a) Hydrogen peroxide decomposes slowly without catalase. Explain how catalase speeds up the decomposition **and** suggest why this is so important in living cells. [2]

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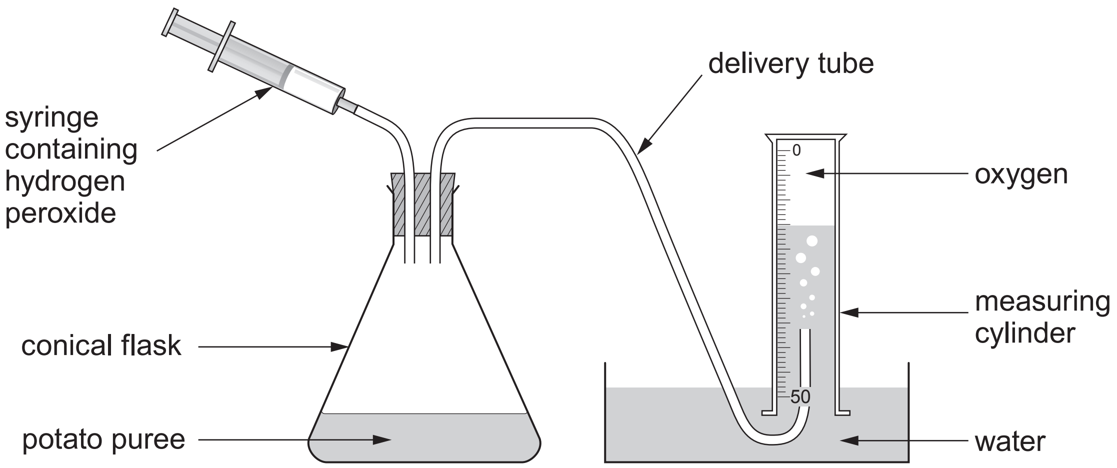
(b) Some students investigated how the concentration of hydrogen peroxide affected the rate of oxygen production. They used fresh potato puree and different concentrations of hydrogen peroxide.
The apparatus was set up as shown in **Image 3.1**.

Image 3.1



- A measuring cylinder was used to measure 20 cm^3 potato puree which was poured into the conical flask.
- A 50 cm^3 measuring cylinder was filled with water and inverted over the open end of the delivery tube.
- 2 cm^3 of 5.0 vol hydrogen peroxide was drawn up into the syringe and attached to the apparatus in the position shown.
- The plunger was pushed in and a timer was started immediately.
- After 30 seconds the volume of oxygen collected was measured using the scale on the measuring cylinder and recorded.
- The rate of oxygen production in $cm^3\ s^{-1}$ was calculated.
- The experiment was repeated using 3.5 vol, 2.0 vol, 1.5 vol, 1.0 vol and 0 vol hydrogen peroxide.

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The results are shown in **Table 3.2**.

Table 3.2

Concentration of hydrogen peroxide / vol	Volume of oxygen collected in 30 seconds / cm^3	Rate of oxygen production / $cm^3\ s^{-1}$
0	0	0.0
1.0	18	0.6
1.5	26	……………………….
2.0	35	1.2
3.5	42	1.4
5.0	43	1.4

(i) Calculate the rate of oxygen production in $cm^3\ s^{-1}$ for **1.5 vol hydrogen peroxide**, **to one decimal place**, and **write your answer in Table 3.2**. [2]

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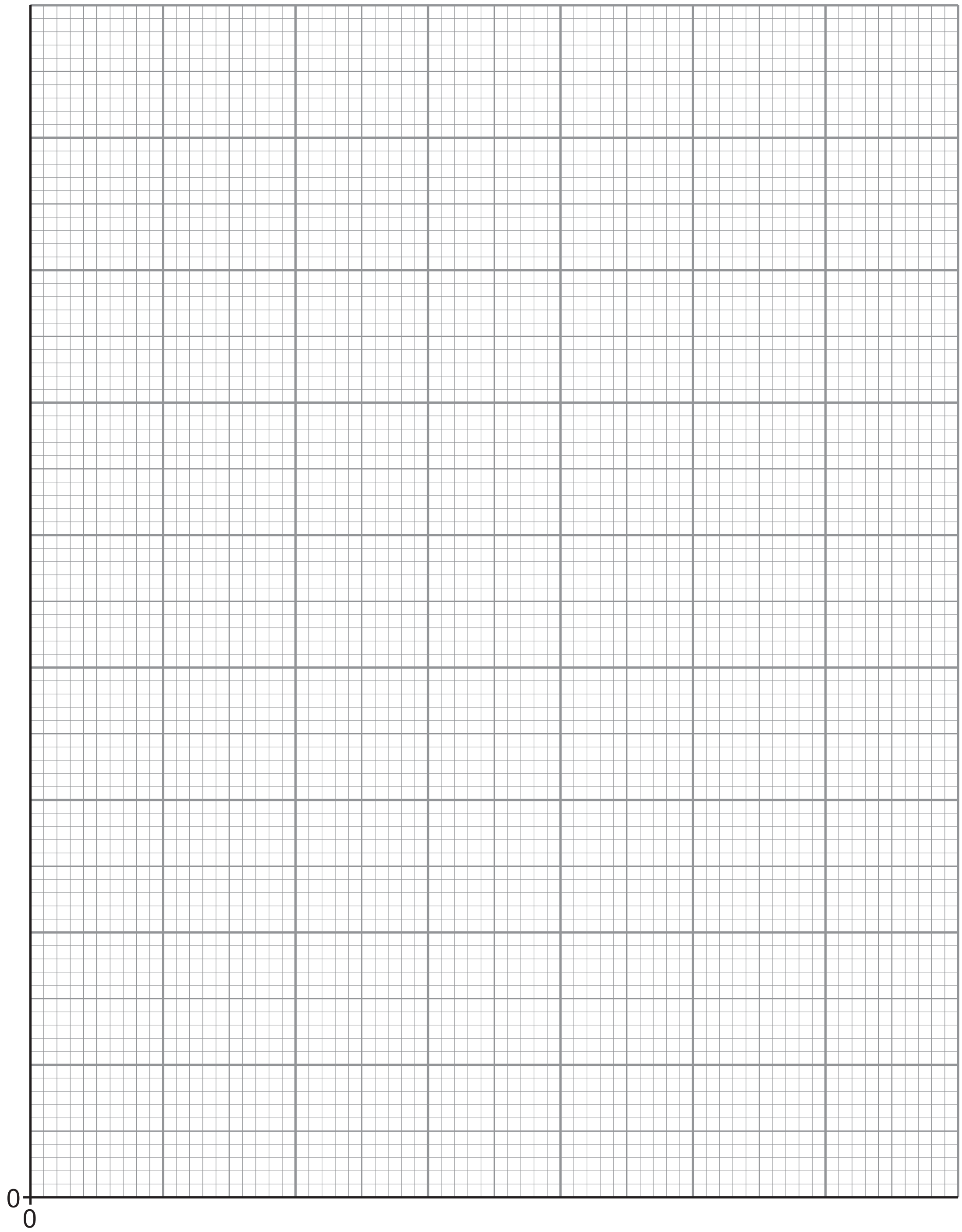
(ii) Use the data from **Table 3.2** to complete **Graph 3.3** to show the rate of oxygen production against the concentration of hydrogen peroxide. [4]

Graph 3.3



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(iii) Describe **and** explain the trend shown in **Graph 3.3**. [3]

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(iv) Suggest **two** ways in which the validity of the data could be improved. [2]

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(v) Copper sulfate is a non-competitive inhibitor of catalase.
On **Graph 3.3**, **draw a labelled line** to show how the result would be different if the experiment was repeated in the presence of copper sulfate. [1]

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